The Economic Value of AI Visibility

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1 Shift-Share IV Construction

Our goal is to estimate the causal effect of AI visibility on brands' sales. Specifically, we are interested in the following model,

$$Sales_{bt} = \alpha_b + \gamma_t + \beta \cdot AIVisibility_{bt} + \varepsilon_{bt} \tag{1}$$

where, $Sales_{bt}$ is the sales of brand b in week t, and $AIVisibility_{bt}$ is the AI visibility of brand b in week t.

The challenge here is that, $AIVisibility_{bt}$ is most likely endogenous, i.e., $\mathbb{E}[\varepsilon_{bt}|AIVisibility_{bt}, \alpha_b, \gamma_t] \neq 0$, because there may be activities by brand, for instance an ad campaign or SEO investment, which may affect both sales and AI visibility. To address this issue, we propose to instrument $AIVisibility_{bt}$ with a Bartik-style shift-share instrument [Bartik, 1991].

The basic formulation of the shift-share IV is the following,

$$B_{bt}^{ssiv} = \sum_k \underbrace{Z_{b0}^k}_{share} \cdot \underbrace{\Gamma_t^k}_{shift} \tag{2}$$

where, k is an entity along which the share and shift are measured. The share is measured in the pre-period, and the shift varies with time.

Here we discuss the formulation and each component of the shift-share IV.

- Unit (b): brand
- Time (t): week
- Entity (k): type of domains (brand page, retail/marketplace, review/news, social media/UGC, reference)
- Shift (Γ_t^k): AI citation share for domain type k in week t
- Share (Z_{b0}^k): share of domain type k in organic search result for keywords containing brand b, in the pre-period

2 Data

Our data comes from multiple sources. We discuss each data source and data collection method below.

2.1 AI Visibility

Our AI visibility measures come from Ahrefs Brand Radar[1]. Brand Radar tracks brand visibility across more than 405 million search-backed prompts and seven AI platforms. The prompts are derived from "People Also Ask" questions in Ahrefs' keyword database and are intended to reflect queries that consumers are likely to search for.

The data provide daily, brand-level AI visibility measures for each brand in our sample from July 1, 2025 through July 31, 2026 (this is for ChatGPT and Gemini; for Google AI Overview and AI Mode, data is available from August 2024). For each brand-day, we observe mention count, brand webpage citation count, impressions and impression-weighted AI share of voice across major AI assistants and AI search interfaces, including ChatGPT, Google AI Overview and AI Mode, and Gemini. We use these data to construct our brand-level AI visibility panel.

[1] https://ahrefs.com/brand-radar

2.2 Domain Citations

Our domain citation data also come from Ahrefs Brand Radar. For each product category, we retain top 20 third-party domains, along with brands' own pages. We collect the day level domain citation data for each domain.

2.3 Transactions

We use the transactions data from ConsumerEdge accessed via the Dewey Data Platform. This is aggregated consumer transaction data on more than 100 million credit and debit cards, from January 1, 2018 through June 27, 2026. We use the data on daily spend breakout on brand and channel type[2]. For our main analysis, we use the online sales data.

2.4 Organic Search Result

We use the Search Rankings[3] data from Serpstat via the Dewey Data Platform. It provides top 50 keyword-level search rankings of nearly 1M website domains in US Google search results.

References

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[2]ConsumerEdge. (2022). Daily Brand and Channel Type (Online/Offline) Breakout [Dataset]. Dewey Data. https://doi.org/10.82551/4CK6-ZT10

[3]Serpstat. (2026). SERP Data [Dataset]. Dewey Data. https://doi.org/10.82551/FZ8W-3767